

Power:

$$P = \alpha C V_{DD}^2 f = C V_{DD}^2 f_{sw}.$$

Dynamic power (switching power)

$$\alpha = \text{Activity factor} = \frac{f_{sw}}{f} \Rightarrow f_{sw} = \alpha f$$

C = Load / output capacitance

V_{DD} = voltage

f = system / clock frequency

$$\alpha = \frac{\text{No. of active transistors}}{\text{No. of total transistors}}$$

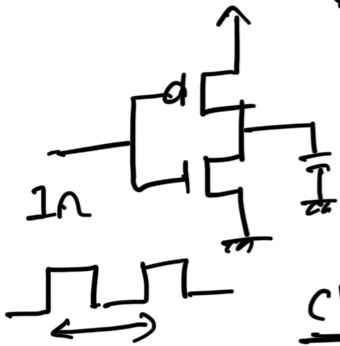
$\Rightarrow$ 10 million transistors. total

1 million " active

$$\alpha = \frac{1}{10} = 0.1$$

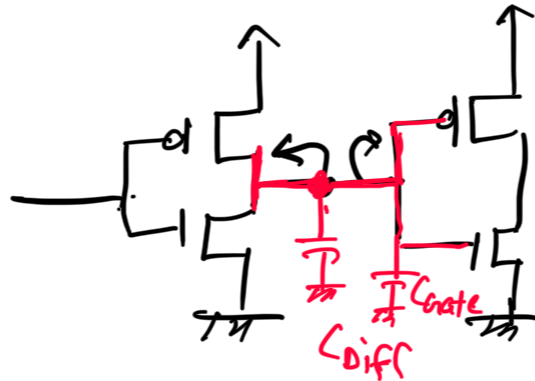
A chip contains 100 million transistor.

Among which 20 million are logic transistors
and the rest 80 million are memory " .

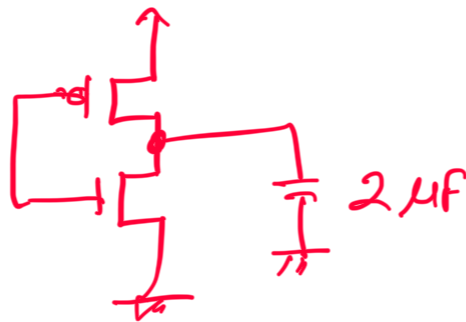


$$\alpha_{Logic} = \frac{20}{100} \times \rightarrow$$

C = Load capacitance



$$C_L = C_{diff} + C_{gate}$$



$$C_{out} = 2 pF$$

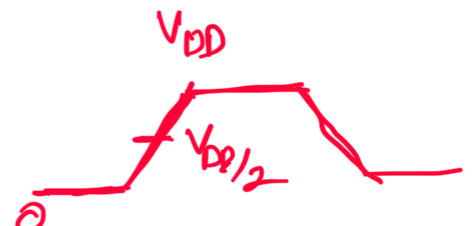
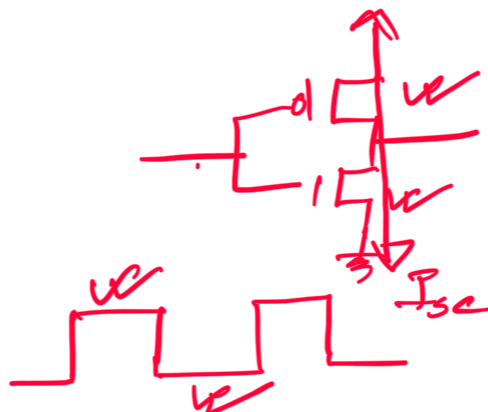
$$C_L = C_{diff} + 2 uF$$

NOR Gate

$$C_{in} = 1 pF$$

$C_T = ?$

$$P_{dynamic} = P_{switching} + P_{short-circuit}$$



Static power

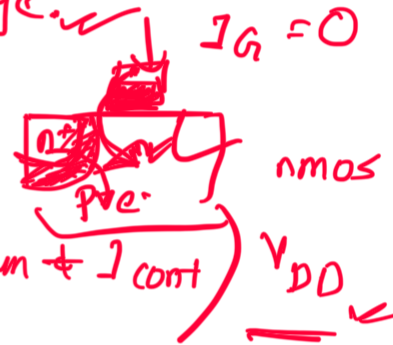
$V_{in} \downarrow / V_{in} \uparrow \rightarrow 0 FF$

1. Subthreshold.

2. Gate leakage.

3. ~~contention~~ Junction leakage.

4. Contention.



$$V_{P_{static}} = (I_{sub} + I_{gate} + I_{jun} + I_{cont}) V_{DD}$$

$$P = V I$$

100 million.

$$I_{sub} = 1 \text{ nA}$$

$$I_{gate} = 1 \text{ pA}$$

$$I_{junction} = 1 \text{ fA}$$

$$P_{static} = (I_{sub} + I_{gate} + I_{junction}) \times V_{DD} \times 100 \text{ million.}$$

each transistor chip

Total transistor = 1 billion

$\lambda = 60 \text{ nm}$, $N_{Logic} = 100 \text{ million}$

$V_{DD} = 0.7 \text{ V}$, $N_{memory} = 700 \text{ million.}$

$f_{sw, Logic} = 1 \text{ GHz}$, $f_{sw, memory} = 3 \text{ GHz}$

$f = 10 \text{ GHz}$, $C_{gate} = 1.2 \text{ fF}/\mu\text{m}$

$C_{diffusion} = 1.7 \text{ fF}/\mu\text{m}$

$$W_{\text{Logic}} = 4 \lambda. \quad W_{\text{memory}} = 5 \lambda.$$

$$P_{\text{gate}} = 10 W. \quad P_{\text{junction}} = 20 W$$

$$P_{\text{subthreshold}} = 5 W,$$

$$P_{\text{short, Logic}} = 50 W$$

~~Σ~~

$$P_{\text{short, memory}} = 70 W$$

$$\alpha_{\text{Logic}} = \frac{1}{10} = 0.1$$

$$\alpha_{\text{memory}} = \frac{3}{10} = 0.3$$

Logic Gate. $\rightarrow V_{DD} = 0.7. \quad W = 4 \lambda = 4 \times 60 \text{ nm}$
 $= 0.24 \text{ } \mu\text{m}$

$$C_L = C_{\text{gate}} + C_{\text{diffusion}}$$

$$= (1.2 + 1.7) \times 10^{-15} \times 0.24 \times 100 \times 10^6$$

$$= 6.96 \times 10^{-8} \text{ F}$$

$$P_{sw} = \alpha C V_{DD}^2 f$$

$$= 0.1 \times 6.96 \times 10^{-8} \times (0.7)^2 \times 10 \times 10^9$$

$$= \boxed{34.104 \text{ W}}$$

Memory. $V_{DD} = 0.7 \quad W = 5 \lambda = 5 \times 60 = 0.3 \text{ } \mu\text{m}.$

$$C_L = (1.2 + 1.7) \times 10^{-15} \times 0.3 \times 700 \times 10^6$$

$$= 6.09 \times 10^{-7} \text{ F}$$

- n a

$$P_{sw} = 0.3 \times 6.09 \times 10^{-7} \times (0.7)^2 \times 10 \times 10^9$$

$$= \boxed{895.23 \text{ W}}$$

$$P_{dynamic} = P_{sw} + P_{short}$$

$$= \underline{34.104 + 895.23 + 60 + 70}$$

$$= 1059.334 \text{ W}$$

$$P_{static} = 10 + 20 + 5 = 35 \text{ W.}$$

$$P_{Total} = P_{static} + P_{dynamic}$$

$$= 1094.334 \text{ W.}$$

$$P_{acceptable} = 1 \text{ kW} = 1000 \text{ W.}$$

$$1094.334 > 1000 \text{ W.} \rightarrow \text{Not acceptable}$$

$$P = \alpha C V_{DD}^2 f \rightarrow V_{DD} \text{ reduce.}$$

Acceptable

$$P_{sw} = 1000 - 60 - 70 - 35$$

$$= 835 \text{ W.}$$

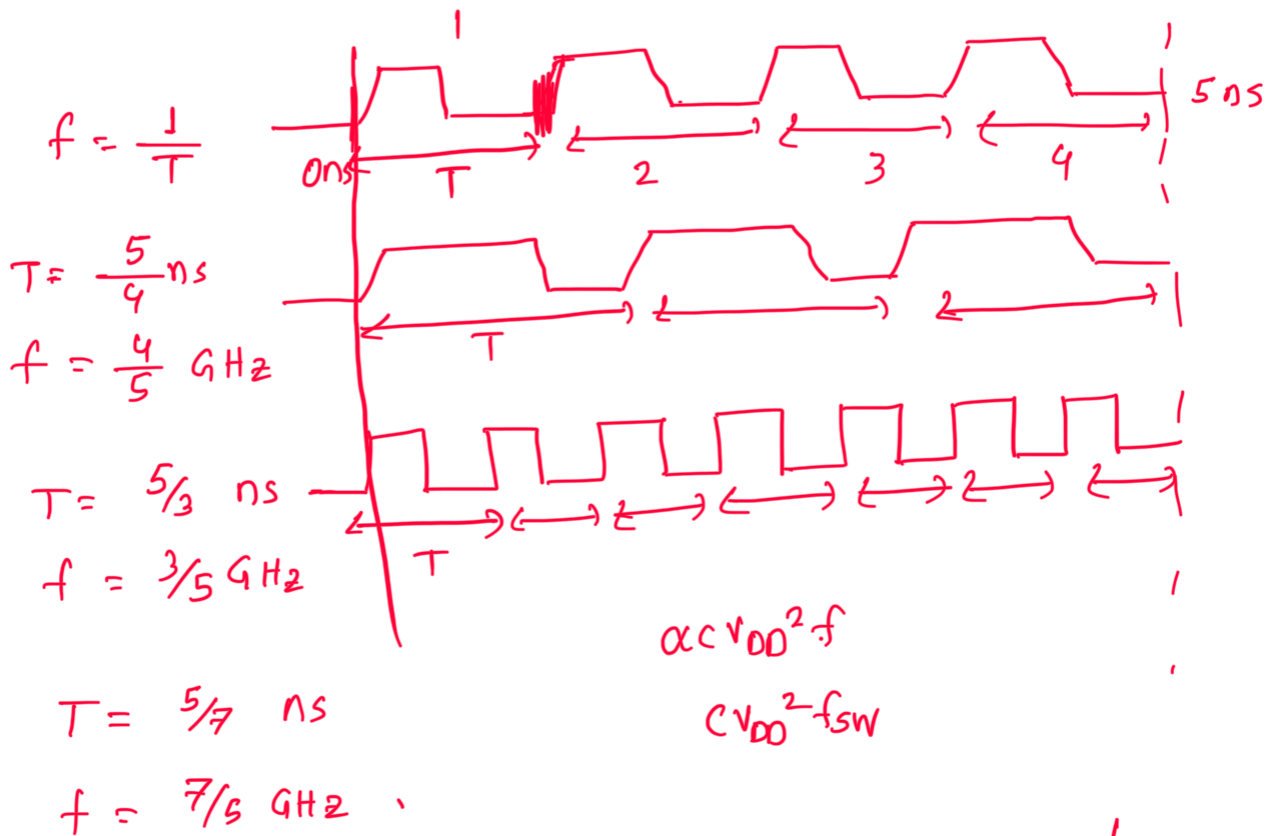
$$P_{sw} = 34.104 + 895.23$$

$$= 929.334 \text{ W.}$$

$$\frac{P_{new}}{P_{old}} = \frac{\cancel{\alpha} \cancel{C} V_{DD, new}^2 \cancel{f}}{\cancel{\alpha} \cancel{C} \cancel{V_{DD, old}^2} \cancel{f}} = \frac{V_{DD, new}^2}{V_{DD, old}^2}$$

$$\begin{aligned}
 V_{DD, new} &= \sqrt{\frac{P_{new}}{P_{old}}} \times V_{DD, old} \\
 &= \sqrt{\frac{835}{929.334}} \times 0.7 \\
 &= 0.6635 \text{ V.}
 \end{aligned}$$

$$\begin{aligned}
 &1 \text{ GHz} \quad 10 \text{ GHz} \quad \frac{1}{10}, \quad \frac{3}{10} \\
 &3 \text{ GHz}
 \end{aligned}$$





shared diffusion ✓

